Operation Lone Star Impact Analysis

Synthetic Difference–in–Differences Analysis of Operation Lone Star's Impact on Border Counties' Employment, Wages, and Establishments

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[GitHub Repository]

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Executive Summary

Operation Lone Star, a program launched by the State of Texas to secure its southern border in March 2021, is associated with reduced economic conditions in the Construction industry (NAICS 23) in Texas counties bordering Mexico when compared to a synthetic control group. Utilizing the most recent Bureau of Labor Statistics data, this study analyzes the 32 Texas border counties defined by the 1983 La Paz Agreement's 100-kilometer (62-mile) international buffer zone. Within this region, the Construction sector's total employment decreased by approximately 20%, average weekly wages by approximately 30%, and business establishments by approximately 12%. None of the other industries tested showed comparable, consistent levels of decline across the three measurements described above.

Operation Lone Star (OLS) utilized state resources – such as the Texas Department of Public Safety, National Guard troops, and physical supplies such as barriers and razor wire – to deter immigrants from entering the state of Texas illegally. The analysis presented in this report utilized the synthetic difference-in-differences (SDiD) model by Arkhangelsky[1] to estimate causal effects between border counties that experienced OLS and border counties that did not. The synthetic control group was built from out of state border counties in New Mexico, Arizona, and California. These donor counties were of similar size and makeup as the treated Texas border counties. This helped the model take into account the economic instability of the COVID-19 epidemic and broader policy changes that may have impacted border counties alongside OLS. This approach helps isolate county-specific changes from broader economic trends that occurred during the same period.

Six industries total were analyzed for impacts of OLS in border counties. Five out of the six industries tested – Agriculture, Wholesale Trade, Transportation & Warehousing, Administrative & Support Services, and Accommodation & Food Services – did not show a statistically significant change in wages or employment. However, total establishment counts declined significantly across all tested industries except for Administrative & Support Services and Wholesale Trade. The absence of a significant employment or wage effect in most industries likely reflects limited statistical power given the small number of treated counties, rather than evidence that OLS had no effect.

It should be noted that this study cannot fully account for the possible impacts of Mexican immigration enforcement, which intensified during the same period as OLS. These efforts from the Mexican government may have impacted economic conditions in border counties alongside OLS. Furthermore, the Quarterly Census, Employment, and Wages (QCEW) data utilized within this study only reflects formally reported jobs and therefore cannot observe the impact of OLS on undocumented labor. These factors may have played an additional role in the border counties' economic conditions but were unable to be captured by current employment reporting methods.

[1] Dmitry Arkhangelsky et al., "Synthetic Difference-in-Differences," American Economic Review 111, no. 12 (December 2021): 4088, https://doi.org/10.1257/aer.20190159.

Key Findings

KEY FINDINGS

- **Construction experienced the most severe downturn** among all examined industries, showing significant declines across all three key metrics: total establishments, average weekly wages, and total employment.
- **In the tested border counties,** a drop in total establishments occurred across four of the six industries, except for Administrative & Support (NAICS 56) and Wholesale Trade (NAICS 42), neither of which reached statistical significance.
- **Beyond these findings,** the remaining industries showed no statistically significant declines in either wages or employment.

Context

Operation Lone Star (OLS) is a Texas state initiative launched in March 2021 by Governor Greg Abbott to deter illegal immigration from occurring along the Texas southern border. The program initiated the deployment of Texas National Guard troops and Department of Public Safety personnel to border counties, leading to the arrest of thousands of migrants who had entered illegally on grounds of state trespassing. The program also provided funding for physical barriers including razor wire and floating buoys in the Rio Grande, which divides Texas and Mexico territories. OLS funding was concentrated in counties where illegal crossings were most prevalent, including Starr, Hidalgo, Maverick, Kinney, and Brooks. However, all Texas border counties were exposed to the economic impacts related to OLS due to changes in available labor and shifting economic conditions. This analysis seeks to analyze how the implementation of OLS meaningfully impacted employment, wages, and business establishments in the Texas border counties where it was implemented. Answering this question may provide useful data regarding economic impacts on local communities for future initiatives.

Data & Methodology

Three indicators were analyzed to measure the possible impact of OLS on border counties' economic conditions: Total number of people employed, their average weekly pay, and how many business establishments were operating per sector. These indicators were measured on a quarterly basis from 2018 through 2025, which is when the most recent data was published by the U.S. Bureau of Labor Statistics [2] at the time of this study. The study analyzed 6 industry sectors, based upon their reliance on the types of labor and trade OLS was likely to disrupt. The sectors include Agriculture, Construction, Wholesale Trade, Transportation & Warehousing, Administrative & Support Services, and Accommodation & Food Services.

The objective of the study was to identify how these sectors may have behaved if OLS had not been implemented and compare it to the real effects of OLS. To accomplish this, a synthetic control group was created based on a weighted combination of counties in California, New Mexico, and Arizona, which share the unique geographic, immigration, and cross-border trade dynamics of the Texas border counties. Although this study utilizes the 1983 La Paz Agreement[3] to define border counties, additional non-border counties were required to adequately populate the donor pool. This is due to Bureau of Labor Statistics (BLS) data suppression in rural jurisdictions, which meant there was an insufficient number of donor counties with complete data to generate the counterfactual. To account for this, the donor pool was expanded to include all county units within the contiguous border states of California, New Mexico, and Arizona. This provided ninety-nine counties available for the donor pool.

However, this does not imply that every donor county was equally weighted; through regularization, SDiD systematically decreases the influence of mismatched inland counties and leverages unit fixed effects to identify parallel growth trends rather than absolute economic scales. The counterfactual was then compared to the real outcomes of the Texas border counties, isolating the treatment effect across the targeted NAICS classifications.

To measure the full effect of OLS, all Texas border counties with unsuppressed Quarterly Census of Employment and Wages data were included in the study. Nine border counties in total were not included in the analysis, due to data suppression at the county level causing the datasets return as empty. This left twenty-three total counties to form the test group.

Limitations

The exclusion of nine Texas border counties due to BLS data suppression introduces a structural constraint regarding statistical power. By reducing the treatment group from thirty-two to twenty-three counties, the model's threshold for detecting policy-induced changes increases. This limitation meant only fairly significant effects would be detectable compared to the control group. Furthermore, several industries displayed sizable declines that did not reach statistical significance in every

[2] U.S. Bureau of Labor Statistics, "Quarterly Census of Employment and Wages," 2018–2025 data panel, accessed August 2026, https://www.bls.gov/cew/.

[3] Agreement on Cooperation for the Protection and Improvement of the Environment in the Border Area (La Paz Agreement), United States–Mexico, August 14, 1983, TIAS no. 10827, https://www.epa.gov/sites/default/files/2015-09/documents/lapazagreement.pdf.

measurement, the most notable being Administrative & Support Services and Wholesale Trade. These measurements likely reflect a lack of statistical power due to a small treatment group rather than evidence that OLS had no effect on those sectors.

It is also worth noting that external factors occurring beyond the state of Texas may have made an impact on border counties' economic conditions. The most notable of these factors includes the COVID-19 pandemic and broader recovery. 2020 through 2021 includes atypical labor shortages and increasing wages, which were especially pronounced in the Construction industry. A second example includes Mexico's own efforts to stem illegal immigration to the US during the same period OLS was implemented, which may have impacted border county's economies separately form OLS. Although the SDiD method was designed to filter out these broader trends, it cannot fully isolate the effects of multiple immigration policies occurring in the same region at the same time.

The total count of hypothesis tests should also be considered. Eighteen total hypothesis tests were run across six industries and three outcomes. Due to this, some individually significant results (such as the establishment declines in agriculture and accommodation/food services) should be interpreted with appropriate caution, as isolated significant findings are more susceptible to chance. The Construction sector's effect is comparatively more credible given its consistency across all three outcome measures rather than resting on a single significant estimate.

One additional aspect worth noting is the data used within the analysis (the Bureau of Labor Statistics' QCEW data) captures only formally reported, legally employed workers. If the effects of OLS occurred primarily in undocumented and unreported labor, it would not directly appear in QCEW data. The declines observed in formal employment, wages, and business counts may understate, or only partially reflect, the policy's full economic footprint in these communities.

Conclusion

The analysis presented above finds that Operation Lone Star is connected with a statistically significant decline in the Construction sector within Texas border counties. These effects were seen across total employment, wages, and total business establishments in the region when compared to the synthetic comparison group. None of the other industries tested showed an effect of similar strength or consistency. While this analysis cannot fully rule out the influence of broader economic events, including Mexico's own enforcement efforts and the COVID-19 pandemic and recovery, the Construction finding holds up across multiple independent methods of defining the affected border region. These findings offer a credible indicator that OLS impacted border counties' economic conditions, primarily within the Construction industry rather than uniformly across all sectors.

Technical Appendix

Data & Methods

Data Source: BLS Quarterly Census of Employment and Wages (QCEW), 2018-2025.

Outcome Variables: Employment (month3_emplvl), average weekly wage (avg_wkly_wage), and establishment count (qtrly_estabs). Each log-transformed (ln(x+1)).

Industries: NAICS 11, 23, 42, 48-49, 56, 72. Selected for probable labor-supply, trade, or logistics disruption due to OLS.

Treatment/Control Groups: Twenty-three Texas border counties were tested against a donor pool of Arizona, New Mexico, and California counties. Treatment coded from 2021 Q2.

Model: Synthetic difference-in-differences[4] estimating regularized unit and time weights via synthdid (R).

Software: Python (pandas) for data pipeline; R (synthdid package) for estimation.

Inference: Standard errors via placebo permutation; 95% CIs as $\hat{\tau} \pm 1.96 \times SE$.

Full Results Tables

Business Establishments

NAICS	Industry	$\hat{\tau}$ (log points)	% Change (point estimate)	95% CI (% change)	Significant?
11	Agriculture	-0.0704	-6.8%	-11.1% to -2.3%	**Yes**
23	Construction	-0.1311	-12.3%	-15.6% to -8.8%	**Yes**
42	Wholesale Trade	-0.0594	-5.8%	-11.3% to +0.2%	No
48-49	Transportation	-0.1255	-11.8%	-17.3% to -6.0%	**Yes**
56	Admin/Support	-0.0237	-2.3%	-7.3% to +2.9%	No
72	Accommodation/Food	-0.0498	-4.9%	-8.4% to -1.1%	**Yes**

Average Weekly Wages

NAICS	Industry	$\hat{\tau}$ (log points)	% Change (point estimate)	95% CI (% change)	Significant?
11	Agriculture	-0.0312	-3.1%	-57.7% to +121.9%	No
23	Construction	-0.3616	-30.3%	-44.8% to -12.1%	**Yes**

[4] Arkhangelsky et al., "Synthetic Difference-in-Differences," 4088.

NAICS	Industry	τ̂ (log points)	% Change (point estimate)	95% CI (% change)	Significant?
42	Wholesale Trade	+0.3302	+39.1%	-31.6% to +182.9%	No
48-49	Transportation	-0.1084	-10.3%	-54.3% to +76.3%	No
56	Admin/Support	-0.1704	-15.7%	-50.0% to +42.3%	No
72	Accommodation/Food	+0.1201	+12.8%	-22.5% to +64.1%	No

Employment Counts

NAICS	Industry	τ̂ (log points)	% Change (point estimate)	95% CI (% change)	Significant?
11	Agriculture	-0.1512	-14.0%	-58.1% to +76.1%	No
23	Construction	-0.2206	-19.8%	-33.5% to -3.3%	**Yes**
42	Wholesale Trade	+0.1552	+16.8%	-39.4% to +125.1%	No
48-49	Transportation	-0.2060	-18.6%	-53.8% to +43.3%	No
56	Admin/Support	-0.3327	-28.3%	-52.6% to +8.5%	No
72	Accommodation/Food	+0.1498	+16.2%	-19.3% to +67.1%	No

County Lists & Exclusions

Treatment group: 23-County (Texas Department of State Health Services Border Region): Brewster (48043), Brooks (48047), Cameron (48061), Crockett (48105), Culberson (48109), Dimmit (48127), Duval (48131), El Paso (48141), Frio (48163), Hidalgo (48215), Hudspeth (48229), Kinney (48271), La Salle (48283), Maverick (48323), Pecos (48371), Presidio (48377), Reeves (48389), Starr (48427), Sutton (48435), Uvalde (48463), Val Verde (48465), Webb (48479), Willacy (48489)

Texas Border Counties Excluded from the Treatment Group:

County	Reason for Exclusion
Edwards	Data suppression
Real	Data suppression
Zavala	Data suppression
Jeff Davis	Data suppression
Terrell	Data suppression
Jim Hogg	Data suppression
Kenedy	Data suppression

County	Reason for Exclusion
McMullen	Data suppression
Zapata	Data suppression

Seven additional control-state counties (in Arizona, New Mexico, and California) were excluded from the donor pool for the same reason.

Code & Reproducibility

The full analysis pipeline is available at this GitHub repository link. The repository is organized as follows:

- /report/ — this report in its source format
- /python/ — data acquisition, cleaning, and panel construction scripts
- /r/ — SDiD estimation script using the synthdid package, looping across all three outcome variables: Employment, establishments, wages

To reproduce the analysis: run the Python pipeline scripts in sequence to acquire and clean the data, then run the corresponding R scripts to generate SDiD estimates for each outcome variable and treatment-group specification.

Bibliography

Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager. "Synthetic Difference-in-Differences." American Economic Review 111, no. 12 (December 2021): 4088–118. https://doi.org/10.1257/aer.20190159.

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